

Chinmay Singhania

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SUMMARY

AI Engineer with hands-on experience in building LLM-powered applications, integrating RAG pipelines, and deploying scalable inference systems using LangChain, OpenAI, and vector databases. Also experienced in building scalable cloud infrastructure, automating CI/CD pipelines using GitHub Actions, and deploying production-grade distributed systems using AWS and Kubernetes.

EDUCATION

North Carolina State University, Raleigh, NC <i>Master of Computer Science</i>	Aug 2023 - May 2025 <i>GPA: 3.89 / 4.0</i>
Savitribai Phule Pune University, Pune, India <i>Bachelor of Engineering in Computer Engineering</i>	Aug 2019 - May 2023 <i>GPA: 8.2 / 10.0</i>

SKILLS & INTERESTS

Programming Languages: Python, Go, Ruby, Java, SQL, Bash, Dart, JavaScript, HTML/CSS
Frameworks: Flask, FastAPI, Streamlit, Ruby on Rails, Flutter, Eclipse
AI/ML & LLM Tools: OpenAI API, LangChain, LLaMA 2, LangFuse, PyTorch, TensorFlow, Keras, Scikit-learn
DevOps & Cloud: Docker, Kubernetes, AWS (EC2, ECS, VPC, ECR, S3, IAM, CloudWatch), Terraform, GitHub Actions
Networking & Storage: TCP/IP, REST, gRPC, MySQL, PostgreSQL, MongoDB, Redis, Apache Cassandra

WORK EXPERIENCE

North Carolina State University <i>Software Engineer</i>	Raleigh, NC <i>Dec 2024 - Present</i>
- Containerized a legacy application to accelerate developer onboarding, reducing local setup time by 50% utilizing Docker. - Migrated CI/CD pipelines from Travis CI to GitHub Actions, implementing automated test workflows for pull requests and commits, resulting in 30% faster pipeline execution and increased reliability. - Promoted to open-source project maintainer, overseeing pull request approvals, code reviews, testing, and smooth integration of community contributions into the production codebase. - Built a full-stack LLM-powered chatbot using React (frontend) and Flask/Python (backend), integrating OpenAI's GPT API to deliver modular conversation flows for virtual teaching and course evaluation. - Implemented retrieval-augmented generation (RAG) by designing optimized MongoDB collections and integrating vector embeddings, enabling intelligent query matching and contextual responses. - Established observability and traceability by integrating Langfuse, enabling detailed tracking of LLM performance and user interactions across multiple conversation threads.	
S2Q2 Software Solutions Pvt. Ltd. (Start up) <i>Software Engineer Intern</i>	Pune, India <i>Jan 2023 - May 2023</i>
- Provisioned scalable AWS infrastructure utilizing Terraform, including VPC, subnets, ECS, and EC2 for containerized deployments. - Automated CI/CD pipeline with GitHub Actions and Ansible, enabling versioned Docker image builds and deployments to ECS. - Integrated CloudWatch dashboards to monitor performance and health, enhancing observability and reducing debugging time.	

PROJECTS

Personalized Blog Generation with LLaMA 2	Project Link
- Designed a content generation app using LLaMA 2 to produce tailored blogs for diverse audiences. - Utilized a quantized model to improve performance and reduce resource consumption for scalable content creation. - Developed a Streamlit-based UI for interactive, real-time content generation and user engagement. - Integrated ctransformers and LangChain for efficient query handling and Sentence-Transformers for relevance optimization in text generation tasks.	
PDF Q&A System with OpenAI + Cassandra	Project Link
- Built a scalable PDF query system using LangChain and Apache Cassandra for efficient document retrieval. - Integrated distributed vector search with DataStax AstraDB for cloud-based high-performance querying. - Implemented natural language query processing with LangChain for precise, dynamic search responses. - Applied OpenAI embeddings to convert PDF text into vectors for similarity-based search. - Optimized vector database operations in Cassandra, ensuring scalability and fault tolerance.	
Scalable Web Microservices	Project Link
- Built scalable RESTful APIs with Go, enabling comparative analysis of monolithic and microservices architectures. - Improved scalability under load by using K8s HPA, achieving optimal resource utilization based on CPU metrics. - Simulated high-traffic with Grafana k6 and visualizing metrics through Prometheus and Grafana dashboards.	

CREDENTIALS

Certified Kubernetes Application Developer (CKAD) by KodeKloud	Certificate Link
Nvidia Getting Started with Deep Learning	Certificate Link
Published Research Paper in IEEE	Paper Link
AWS Fundamentals: Addressing Security Risk	Certificate Link
AWS Fundamentals: Going Cloud-Native	Certificate Link